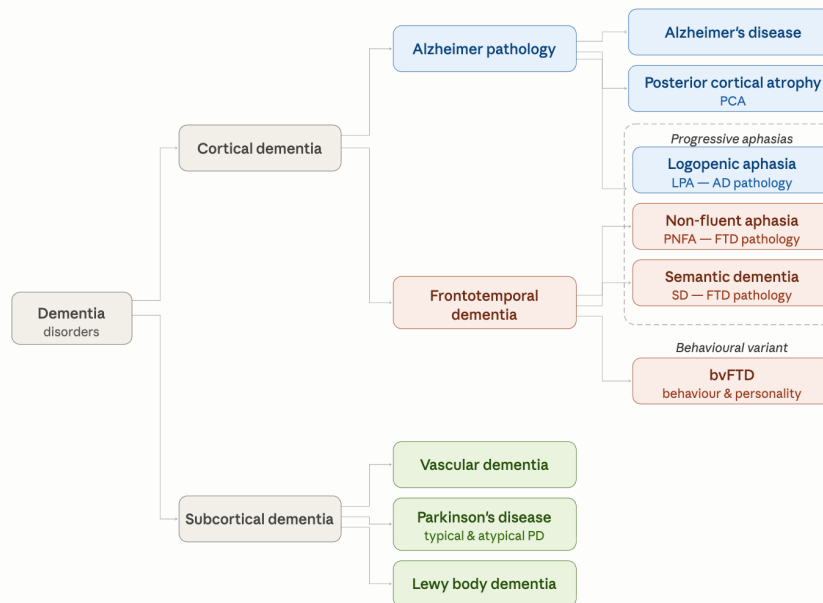




Dementia

is an umbrella term used to describe a collection of brain diseases and their symptoms including: memory loss, impaired judgement, personality changes etc. severe enough to impair daily functioning



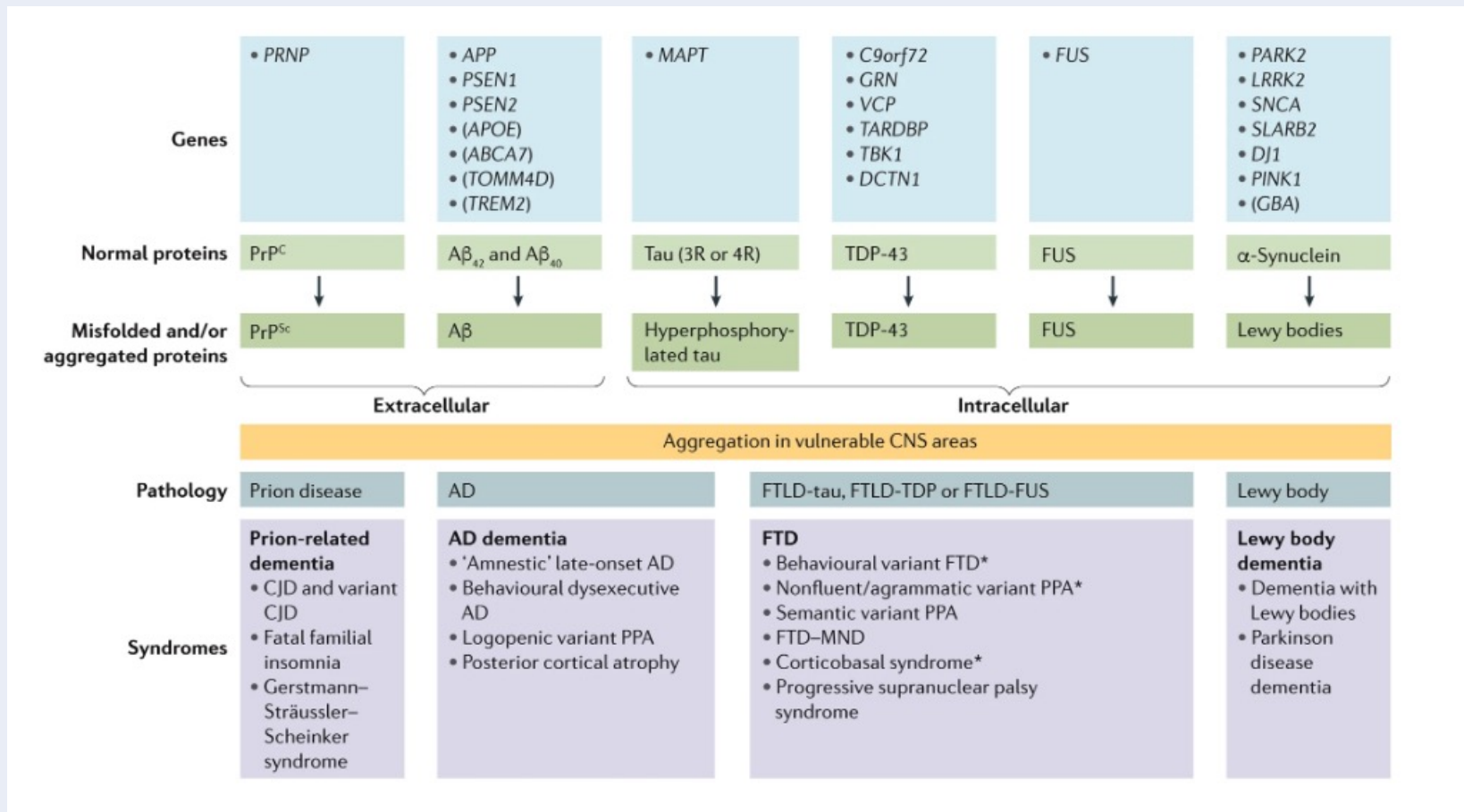


1. Dementia Epidemiology & Definition

MCI vs. Dementia

- **Dementia:** acquired, persistent cognitive decline in ≥ 1 domain (memory, language, executive function, visuospatial, social cognition) — severe enough to impair everyday functioning
- **MCI (Mild Cognitive Impairment):** objective decline beyond normal ageing — but independence in daily activities preserved; intermediate zone between normal ageing and dementia
- **Conversion rate:** ~10–15% of MCI patients per year progress to dementia — depends on aetiology and biomarker profile
- dementia is defined by functional impairment, not by test scores alone

Summary of proteinopathies



Elahi & Miller, 2017 in Rev Neurol



Summary of clinical profiles:

Type	First symptom	Memory	Behaviour	Motor / Other
Alzheimer's	Episodic memory loss	Severe, early	Late stage	Late; Apraxia, Gait disturbance
Vascular	Executive / attention	Moderate	Variable	Gait, incontinence
LBD	Visuospatial / attention	Moderate	REM sleep disorder early	Parkinsonism, hallucinations
FTD (bvFTD)	Personality / behaviour	Preserved early	Severe, early	Late; ALS overlap possible



3. Behavioral Symptoms of Dementia

- Up to 90% of patients experience BPSD at some point
- Primary driver of caregiver burnout, hospitalisation, and nursing home placement
- Always assess for underlying physical causes first (pain, infection, constipation, fear)

Apathy

50–70%; reduced motivation & emotional responsiveness — distinct from depression

Agitation / Aggression

Often triggered by unmet needs; assess for pain before medicating

Psychosis

Visual hallucinations (DLB); paranoid delusions in AD (theft, infidelity)

Depression

30–40%; late-onset depression itself is an AD risk factor

Sleep disturbances

Sundowning; REM- sleep disorder for α -synucleinopathies

Anosognosia

30–50% in AD; unawareness → risky behaviour (driving, finances)



4. Differential diagnostics

- **Depression vs. AD:** subjective > objective deficit; variable performance; biomarkers negative; mood symptoms prominent
- **LBD vs. AD:** fluctuation, visual hallucinations, parkinsonism, RBD — DaT-SPECT abnormal; neuroleptic sensitivity!
- **FTD vs. AD:** younger onset; behaviour precedes memory; neuropsych: executive > memory; frontal atrophy
- **Vascular vs. AD:** vascular risk factors; stepwise course; MRI white matter changes; executive dysfunction profile
- **Delirium vs. Dementia:** acute onset, fluctuating consciousness, impaired attention — always exclude acute medical cause first
- **Reversible causes:** NPH, hypothyroidism, B12, autoimmune encephalitis — structural imaging mandatory



4. Differential diagnostics

■ Depression vs. AD

Cognitive complaints

Test vs. daily performance

Cognitive profile

Affect

Response to antidepressant treatment

Biomarkers

Alzheimer's Disease

Symptoms tend to be minimised or denied

Test performance and everyday functioning are consistent

Typical profile: memory consolidation predominantly affected; visuospatial deficits emerge early

Early stages often depressive; apathy predominates over sadness; moderate stages: balanced or dysphoric-irritable

No improvement in cognitive symptoms with antidepressive therapy

Abnormal biomarkers (CSF, MRI, PET, SPECT)

Depression

Prominent complaints about cognitive deficits, detailed and elaborated

Discrepancy between test performance and reported daily functioning ("I just can't do it")

Non-specific profile: attention and cognitive flexibility most affected; visuospatial or aphasic symptoms almost never present

Pronounced depressive affect, early morning awakening, rumination, self-doubt

Cognitive symptoms may improve with successful antidepressive therapy (residual deficits possible in older patients)

Biomarkers unremarkable



6. Prevention

- **Lancet Commission 2024:** ~45% of dementia cases are attributable to 14 modifiable risk factors
- **Early life:** low education
- **Midlife:** hearing loss, hypertension, obesity, smoking, depression, physical inactivity, diabetes, excessive alcohol, traumatic brain injury
- **Later life:** air pollution, social isolation, untreated vision loss, high LDL (new 2024)
- **Key message:** dementia is not inevitable — targeting these factors could prevent or delay nearly half of all cases



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